InvokeAgent

Service: Agents for Amazon Bedrock Runtime

> **Note**
>
> The AWS CLI doesn't support streaming operations in Amazon Bedrock, including `InvokeAgent`.

Sends a prompt for the agent to process and respond to. Note the following fields for the request:

- To continue the same conversation with an agent, use the same `sessionId` value in the request.
- To activate trace enablement, turn `enableTrace` to `true`. Trace enablement helps you follow the agent's reasoning process that led it to the information it processed, the actions it took, and the final result it yielded. For more information, see Trace enablement.
- End a conversation by setting `endSession` to `true`.
- In the `sessionState` object, you can include attributes for the session or prompt or, if you configured an action group to return control, results from invocation of the action group.

The response contains both **chunk** and **trace** attributes.

The final response is returned in the `bytes` field of the `chunk` object. The `InvokeAgent` returns one chunk for the entire interaction.

- The `attribution` object contains citations for parts of the response.
- If you set `enableTrace` to `true` in the request, you can trace the agent's steps and reasoning process that led it to the response.
- If the action predicted was configured to return control, the response returns parameters for the action, elicited from the user, in the `returnControl` field.
- Errors are also surfaced in the response.

Request Syntax

```
POST /agents/agentId/agentAliases/agentAliasId/sessions/sessionId/text HTTP/1.1
x-amz-source-arn: sourceArn
Content-type: application/json
```

```
{
   "bedrockModelConfigurations": {
      "performanceConfig": {
         "latency": "string"
      }
   },
   "enableTrace": boolean,
   "endSession": boolean,
   "inputText": "string",
   "memoryId": "string",
   "promptCreationConfigurations": {
      "excludePreviousThinkingSteps": boolean,
      "previousConversationTurnsToInclude": number
   },
   "sessionState": {
      "conversationHistory": {
         "messages": [
            {
               "content": [
                  { ... }
               ],
               "role": "string"
            }
         ]
      },
      "files": [
         {
            "name": "string",
            "source": {
               "byteContent": {
                  "data": blob,
                  "mediaType": "string"
               },
               "s3Location": {
                  "uri": "string"
               },
               "sourceType": "string"
            },
            "useCase": "string"
         }
      ],
      "invocationId": "string",
      "knowledgeBaseConfigurations": [
         {
```

```
            "knowledgeBaseId": "string",
            "retrievalConfiguration": {
               "vectorSearchConfiguration": {
                  "filter": { ... },
                  "implicitFilterConfiguration": {
                     "metadataAttributes": [
                        {
                           "description": "string",
                           "key": "string",
                           "type": "string"
                        }
                     ],
                     "modelArn": "string"
                  },
                  "numberOfResults": number,
                  "overrideSearchType": "string",
                  "rerankingConfiguration": {
                     "bedrockRerankingConfiguration": {
                        "metadataConfiguration": {
                           "selectionMode": "string",
                           "selectiveModeConfiguration": { ... }
                        },
                        "modelConfiguration": {
                           "additionalModelRequestFields": {
                              "string" : JSON value
                           },
                           "modelArn": "string"
                        },
                        "numberOfRerankedResults": number
                     },
                     "type": "string"
                  }
               }
            }
         }
      ],
      "promptSessionAttributes": {
         "string" : "string"
      },
      "returnControlInvocationResults": [
         { ... }
      ],
      "sessionAttributes": {
         "string" : "string"
```

```
        }
    },
    "streamingConfigurations": {
        "applyGuardrailInterval": number,
        "streamFinalResponse": boolean
    }
}
```

URI Request Parameters

The request uses the following URI parameters.

agentAliasId

The alias of the agent to use.

Length Constraints: Minimum length of 0. Maximum length of 10.

Pattern: `[0-9a-zA-Z]+`

Required: Yes

agentId

The unique identifier of the agent to use.

Length Constraints: Minimum length of 0. Maximum length of 10.

Pattern: `[0-9a-zA-Z]+`

Required: Yes

sessionId

The unique identifier of the session. Use the same value across requests to continue the same conversation.

Length Constraints: Minimum length of 2. Maximum length of 100.

Pattern: `[0-9a-zA-Z._:-]+`

Required: Yes

sourceArn

The ARN of the resource making the request.

Length Constraints: Minimum length of 0. Maximum length of 2048.

Pattern: `arn:aws(-[^:]+)?:bedrock:[a-z0-9-]{1,20}:[0-9]{12}:agent/[0-9a-zA-Z]{10}`

Request Body

The request accepts the following data in JSON format.

bedrockModelConfigurations

Model performance settings for the request.

Type: BedrockModelConfigurations object

Required: No

enableTrace

Specifies whether to turn on the trace or not to track the agent's reasoning process. For more information, see Trace enablement.

Type: Boolean

Required: No

endSession

Specifies whether to end the session with the agent or not.

Type: Boolean

Required: No

inputText

The prompt text to send the agent.

> **Note**
>
> If you include `returnControlInvocationResults` in the `sessionState` field, the `inputText` field will be ignored.

Type: String

Length Constraints: Minimum length of 0. Maximum length of 25000000.

Required: No

memoryId

The unique identifier of the agent memory.

Type: String

Length Constraints: Minimum length of 2. Maximum length of 100.

Pattern: `[0-9a-zA-Z._:-]+`

Required: No

promptCreationConfigurations

Specifies parameters that control how the service populates the agent prompt for an `InvokeAgent` request. You can control which aspects of previous invocations in the same agent session the service uses to populate the agent prompt. This gives you more granular control over the contextual history that is used to process the current request.

Type: PromptCreationConfigurations object

Required: No

sessionState

Contains parameters that specify various attributes of the session. For more information, see Control session context.

> **Note**
>
> If you include `returnControlInvocationResults` in the `sessionState` field, the `inputText` field will be ignored.

Type: SessionState object

Required: No

streamingConfigurations

Specifies the configurations for streaming.

> **Note**
>
> To use agent streaming, you need permissions to perform the `bedrock:InvokeModelWithResponseStream` action.

Type: StreamingConfigurations object

Required: No

Response Syntax

```
HTTP/1.1 200
x-amzn-bedrock-agent-content-type: contentType
x-amz-bedrock-agent-session-id: sessionId
x-amz-bedrock-agent-memory-id: memoryId
Content-type: application/json

{
   "accessDeniedException": {
   },
   "badGatewayException": {
   },
   "chunk": {
      "attribution": {
         "citations": [
            {
               "generatedResponsePart": {
                  "textResponsePart": {
                     "span": {
                        "end": number,
                        "start": number
                     },
                     "text": "string"
                  }
               },
               "retrievedReferences": [
                  {
                     "content": {
                        "audio": {
                           "s3Uri": "string",
                           "transcription": "string"
```

```
                    },
                    "byteContent": "string",
                    "row": [
                        {
                            "columnName": "string",
                            "columnValue": "string",
                            "type": "string"
                        }
                    ],
                    "text": "string",
                    "type": "string",
                    "video": {
                        "s3Uri": "string",
                        "summary": "string"
                    }
                },
                "location": {
                    "confluenceLocation": {
                        "url": "string"
                    },
                    "customDocumentLocation": {
                        "id": "string"
                    },
                    "kendraDocumentLocation": {
                        "uri": "string"
                    },
                    "s3Location": {
                        "uri": "string"
                    },
                    "salesforceLocation": {
                        "url": "string"
                    },
                    "sharePointLocation": {
                        "url": "string"
                    },
                    "sqlLocation": {
                        "query": "string"
                    },
                    "type": "string",
                    "webLocation": {
                        "url": "string"
                    }
                },
                "metadata": {
```

```
                        "string" : JSON value
                    }
                }
            ]
        }
    ]
  },
  "bytes": blob
},
"conflictException": {
},
"dependencyFailedException": {
},
"files": {
  "files": [
    {
      "bytes": blob,
      "name": "string",
      "type": "string"
    }
  ]
},
"internalServerException": {
},
"modelNotReadyException": {
},
"resourceNotFoundException": {
},
"returnControl": {
  "invocationId": "string",
  "invocationInputs": [
    { ... }
  ]
},
"serviceQuotaExceededException": {
},
"throttlingException": {
},
"trace": {
  "agentAliasId": "string",
  "agentId": "string",
  "agentVersion": "string",
  "callerChain": [
    { ... }
```

```
        ],
        "collaboratorName": "string",
        "eventTime": "string",
        "sessionId": "string",
        "trace": { ... }
    },
    "validationException": {
    }
}
```

Response Elements

If the action is successful, the service sends back an HTTP 200 response.

The response returns the following HTTP headers.

contentType

The MIME type of the input data in the request. The default value is `application/json`.

memoryId

The unique identifier of the agent memory.

Length Constraints: Minimum length of 2. Maximum length of 100.

Pattern: `[0-9a-zA-Z._:-]+`

sessionId

The unique identifier of the session with the agent.

Length Constraints: Minimum length of 2. Maximum length of 100.

Pattern: `[0-9a-zA-Z._:-]+`

The following data is returned in JSON format by the service.

accessDeniedException

The request is denied because of missing access permissions. Check your permissions and retry your request.

Type: Exception

HTTP Status Code: 403

badGatewayException

There was an issue with a dependency due to a server issue. Retry your request.

Type: Exception

HTTP Status Code: 502

chunk

Contains a part of an agent response and citations for it.

Type: PayloadPart object

conflictException

There was a conflict performing an operation. Resolve the conflict and retry your request.

Type: Exception

HTTP Status Code: 409

dependencyFailedException

There was an issue with a dependency. Check the resource configurations and retry the request.

Type: Exception

HTTP Status Code: 424

files

Contains intermediate response for code interpreter if any files have been generated.

Type: FilePart object

internalServerException

An internal server error occurred. Retry your request.

Type: Exception

HTTP Status Code: 500

modelNotReadyException

The model specified in the request is not ready to serve Inference requests. The AWS SDK will automatically retry the operation up to 5 times. For information about configuring automatic retries, see Retry behavior in the *AWS SDKs and Tools* reference guide.

Type: Exception

HTTP Status Code: 424

resourceNotFoundException

The specified resource Amazon Resource Name (ARN) was not found. Check the Amazon Resource Name (ARN) and try your request again.

Type: Exception

HTTP Status Code: 404

returnControl

Contains the parameters and information that the agent elicited from the customer to carry out an action. This information is returned to the system and can be used in your own setup for fulfilling the action.

Type: ReturnControlPayload object

serviceQuotaExceededException

The number of requests exceeds the service quota. Resubmit your request later.

Type: Exception

HTTP Status Code: 400

throttlingException

The number of requests exceeds the limit. Resubmit your request later.

Type: Exception

HTTP Status Code: 429